AI LAB

WEEK 4

23/10/26

BATCH 2

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Hill Climbing (N-Queens)

CODE:

```
# Hill Climbing for 4-Queens

def cost(state):
    conflicts = 0

    for i in range(4):
        for j in range(i + 1, 4):

            if state[i] == state[j]:
                conflicts += 1

            elif abs(state[i] - state[j]) == abs(i - j):
                conflicts += 1

    return conflicts


def hill_climbing():
    current = [3, 1, 2, 0]

    while True:
        print("Current state:", current)
        print("Current cost:", cost(current))

        best = current[:]

        # Try swapping every pair of queens
        for i in range(4):
            for j in range(i + 1, 4):
                neighbor = current[:]

                neighbor[i], neighbor[j] = neighbor[j], neighbor[i]

                print("Neighbor:", neighbor,
                      "Cost:", cost(neighbor))

                if cost(neighbor) < cost(best):
                    best = neighbor

        # Stop if no better neighbor exists
        if cost(best) >= cost(current):
            break
```

```
        current = best
        print()

    print("\nFinal state:", current)
    print("Final cost:", cost(current))

    if cost(current) == 0:
        print("Solution found!")


hill_climbing()
```

OUTPUT:

```
Current state: [3, 1, 2, 0]
Current cost: 2
Neighbor: [1, 3, 2, 0] Cost: 1
Neighbor: [2, 1, 3, 0] Cost: 1
Neighbor: [0, 1, 2, 3] Cost: 6
Neighbor: [3, 2, 1, 0] Cost: 6
Neighbor: [3, 0, 2, 1] Cost: 1
Neighbor: [3, 1, 0, 2] Cost: 1

Current state: [1, 3, 2, 0]
Current cost: 1
Neighbor: [3, 1, 2, 0] Cost: 2
Neighbor: [2, 3, 1, 0] Cost: 2
Neighbor: [0, 3, 2, 1] Cost: 4
Neighbor: [1, 2, 3, 0] Cost: 4
Neighbor: [1, 0, 2, 3] Cost: 2
Neighbor: [1, 3, 0, 2] Cost: 0

Current state: [1, 3, 0, 2]
Current cost: 0
Neighbor: [3, 1, 0, 2] Cost: 1
Neighbor: [0, 3, 1, 2] Cost: 1
Neighbor: [2, 3, 0, 1] Cost: 4
Neighbor: [1, 0, 3, 2] Cost: 4
Neighbor: [1, 2, 0, 3] Cost: 1
Neighbor: [1, 3, 2, 0] Cost: 1

Final state: [1, 3, 0, 2]
Final cost: 0
Solution found!
```